Seat No.: ________ Enrolment No. ________

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (December – 2024)

M.Sc. Cyber Security Semester – I

Subject Code: CTMSCS-SI P5 **Date: 10/12/2024**

Subject Name: Introduction to Forensic Science and Cyber Law

Time: 2:30pm – 5:30pm **Total Marks: 100**

Instructions:

1. Write down each question on a separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

			Marks
Q.1		**Attempt any three.**	
	(a)	Explain ethics in Forensic Sciences	08
	(b)	Write about the role of forensic scientist	08
	(c)	Write a detail note on Microscopy	08
	(d)	Explain the Organizational Setup of Forensic Science in India	08
Q.2		**Attempt any three.**	
	(a)	Explain Section 375(IPC) or Sec 63(BNS) in Detail	08
	(b)	Write a note on GEQD and BPR&D	08
	(c)	Write about the history of Forensic Science.	08
	(d)	Explain Section 45 and 46 of IEA	08
Q.3		**Attempt any three.**	
	(a)	Write detail note on major instrumentation used in Forensic laboratory	08
	(b)	Explain IT Act.	08
	(c)	Explain Section 299(IPC) or Sec 100 (BNS) in detail	08
	(d)	Explain the hierarchy of courts in India and their powers in detail	08
Q.4		**Attempt any two.**	
	(a)	Explain Different branches of forensic sciences in detail	07
	(b)	Explain the Need of Forensic Science.	07
	(c)	Define Computer, its components and different types of storage media.	07
Q.5		**Attempt any two.**	
	(a)	Explain 7 laws of Forensic Science.	07
	(b)	Explain the working of Video Spectral Comparator and ESDA	07
	(c)	Write a Brief note on NIA and CCNTS	07

--- End of Paper---

National Forensics Sciences University, Goa Campus
TA-1 Examination

Program Name – MSc. Cyber/DFIS/MTech	Sem – I	Date- 10/09/25
Subject Name- Introduction to Forensic Science & cyber law		
Subject Code- MSCS/DFIS/M.Tech-SI-P5		
Time- 45 minutes		Max. Marks- 25

A		**Short Answer Questions**	**15 Marks**
	I.	What do you understand by Data Depiction?	2 Marks
	II.	Draw a flow chart of the organizational set-up of forensic science laboratories in India.	2 Marks
	III.	Name the CFSLs present in India.	2 Marks
	IV.	Name the different divisions where a blood sample containing drugs will be sent and mention the 2 methods of analysis.	2 Marks
	V.	Name the evidence these divisions deal with - a) Anthropometric- b) Serology- c) Odontology- d) Cyber Forensic	2 Marks
	VI.	What are the duties of Forensic Scientists?	2.5 Marks
	VII.	Briefly write about the code of conduct of a forensic scientist in Forensic science. (4 lines)	2.5 Marks
B		**Answer any 2 questions (2 x 5 marks each)**	**10 Marks**
	I.	Prepare a comprehensive report on the digital evidence given to you.	5 Marks
	II.	Discuss the various laws of Forensic Science with suitable examples.	5 marks
	III.	Why Forensic science is a multi-disciplinary field? And what is its significance?	5 Marks

All the best

National Forensics Sciences University, Goa Campus
TA-1 Examination

Program Name – M.Sc. Cyber Security Date- 09/09/25 Subject Name- Artificial Intelligence Time- 45 minutes		Sem – I Subject Code- (CTMSCS SI P4) Max. Marks- 25
I	**Short Answer Questions**	**10 marks**
	I. Differentiate between a Python module and a package, detailing their creation and import mechanisms. Additionally, explain the role and functionality of context processors in Python-based web frameworks.	2 marks
	II. Design a Python class called DataProcessor that encapsulates statistical operations. The class should have methods to calculate mean, median, and standard deviation of a dataset. Include a class method to create an instance from a list and demonstrate its usage with sample data.	2 marks
	III. Develop a Python snippet that constructs a dictionary mapping student name to their scores, ensuring typecasting from strings to integers and validating that each score falls within an acceptable range of 0-100. Include at least three entries.	2 marks
	IV. What is a data class in Python? Mention one benefit of using data class over a regular class with example.	2 marks
	V. List two key differences between NumPy arrays and Python lists. Focus on performance and functionality.	2 marks
II	**Answer any 3 questions (3 x 5 marks each)**	**15 Marks**
	i. Explain how to set up a Python development environment using Anaconda, including creating a new environment, installing NumPy and Pandas, and computing the cube of the matrix [1, 0, -1; 1, 0, 0; 1, -1, 2].	5 marks
	ii. the role of loops and conditional statements in Python programming. Write a script that takes a list of integers and prints whether each number is even or odd, using both loop and condition, also time the script using decorators.	5 marks
	iii. Describe how Matplotlib and SciPy can be used together for scientific visualization and computation. Provide an example where you plot a sine wave using Matplotlib and compute its sum of series over interval [π/12, π/3].	5 marks
	iv. List all bugs in following program: import maths def compute_circle_area(radius) if radius <= 0:	5 marks

	`` print("Radius must be positive") return area = math.pi * radius ^ 2 return area def display_result(area): print("The area of the circle is: " + area) radii = [5, -3, 'seven', None, 0, 3.5, [2], True] for r in radii: result = compute_circle_area(r) display_result(result) print("All areas computed successfully") ```	

All the best

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Seat No.: ____________ Enrolment No. ____________

NATIONAL FORENSIC SCIENCES UNIVERSITY
M.SC. CYBER SECURITY

Subject Code: CTMSCS SI P4
Subject Name: Artificial Intelligence
Time: 02:30 PM to 05:30 PM

Date: 9/12/2024

Total Marks: 100

Instructions:
1. **Write down each question on separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

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			Marks
Q.1		**Attempt any three.**	
	(a)	Create a Python function analyze_list that takes a list of integers and returns a dictionary with keys 'sum', 'average', 'max', and 'min'. Demonstrate this function with the list [10, 20, 30, 40, 50].	4+4
	(b)	i. Imagine you are in a grocery store. You want to buy 1 rose, 2 masalas, and 3 milk bottles. The unit prices are $1, $2, $0.5. respectively. Generate Dot products of given vector. ii. You have a dataset representing the daily temperatures in degrees Celsius for a week: Temperatures = [20,22,21,25,24]. Calculate the variance of the temperatures for the week.	3+5
	(c)	Using NumPy, create a 1D array of numbers from 0 to 9, reshape it into a 2x5 array, create a DataFrame from the reshaped array using Pandas, and plot a bar chart of the DataFrame using Matplotlib.	1+2+2+3
	(d)	Write a short on .describe( ) function in python, with an example. Compare it with .info( ) and .shape( ) commands in python	4+2+2
Q.2		**Attempt any three.**	
	(a)	Explain the steps to set up a Python development environment using Anaconda and write a Python script that takes a user input number and checks if it is even or odd, printing "Even" or "Odd" accordingly.	08
	(b)	Explain KNN algorithm and find the Euclidian distance between new customers named 'Monica' has height 134cm and weight 64 kg and old customer name 'Rama' has height 123 cm and weight 57kg.	08
	(c)	Describe the process of splitting a dataset into training, testing, and validation sets. How does the use of validation set help in model evaluation and hyperparameter tuning? Provide an example to illustrate your answer.	2+6
	(d)	Explain working of K-means & Hierarchical clustering. Discuss about pros and cons of both unsupervised algorithms.	6+2

<table>
<tr><td>Q.3</td><td></td><td>Attempt any three.</td><td></td></tr>
<tr><td></td><td>(a)</td><td>Using NumPy, create a 1D array of numbers from 0 to 9, reshape it into a 2x5 array, create a DataFrame from the reshaped array using Pandas, and plot a bar chart of the DataFrame using Matplotlib.</td><td>1+2+2+3</td></tr>
<tr><td></td><td>(b)</td><td>a) Discuss two major issues and challenges in Machine Learning.
b) Explain the different types of Machine Learning and provide an example of each type.</td><td>4+4</td></tr>
<tr><td></td><td>(c)</td><td>Draw and describe ANN Architecture and CNN Architecture and write the difference between them.</td><td>3+5</td></tr>
<tr><td></td><td>(d)</td><td>i. Obtain and draw the confusion matrix for the following python lines
Consider the True values
y_true_S=np.array([H1,H1,H1,H1,H1,H1,H1,H1,H1,H1,H1,H1,H1,
H1,H1,H1,H1,H1,H1,H1,H2,H2,H2,H2,H2,H2,H2,H2,H2,H2,H2,H2
,H2,H2,H2,H2,H2,H2,H2,H2])
Consider the Predicted values
y_pred_S=np.array([H1,H1,H1,H1,H1,H1,H1,H1,H1,H1,H1,H1,H1
,H1,H1,H1,H1,H2,H2,H2,H1,H1,H2,H2,H2,H2,H2,H2,H2,H2,H2,H
2,H2,H2,H2,H2,H2,H2,H2,H2])
ii. From the given matrix, write the meaning of each of 4 values.
<table><tr><td>51</td><td>16</td></tr><tr><td>21</td><td>27</td></tr></table>x-axis (non-event and event) and y-axis (non-event and event) are actual and predicted values.
iii. With the help of a neat diagram, justify: Increasing the precision of a ML model, decreases the recall, and vice-versa. State and explain the metric: F1-score.</td><td>2+2+4</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td>Q.4</td><td></td><td>Attempt any two.</td><td></td></tr>
<tr><td></td><td>(a)</td><td>What are the different techniques/steps involved in Detecting and Recognizing Faces   OB FE GOK Analysis Post Pro</td><td>07</td></tr>
<tr><td></td><td>(b)</td><td>What are the steps involved in Natural Language Pipeline. Explain any four with example.</td><td>07</td></tr>
<tr><td></td><td>(c)</td><td>Draw and Explain the LSTM Architecture.</td><td>07</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td>Q.5</td><td></td><td>Attempt any two.</td><td></td></tr>
<tr><td></td><td>(a)</td><td>How machine learning can be used to detect phishing attacks and identify fake content.</td><td>07</td></tr>
<tr><td></td><td>(b)</td><td>You are tasked with designing an ML-based cyber security system for a large enterprise. The system needs to address multiple aspects of security, including malware detection, anomaly detection, and intrusion detection. Describe the design of your system.</td><td>3+4</td></tr>
<tr><td></td><td>(c)</td><td>Discuss how ML can be applied in anomaly detection and penetration testing (Pen Testing) to improve the security of systems and networks.</td><td>07</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
</table>

--- End of Paper---

National Forensics Sciences University, Goa Campus
Mid-semester Examination

Programme – M.Sc. Cyber Security	Sem – I		Date- 10.10.2024
Subject Name- Web Application Security	Subject Code- CTMSCS SI P3		
Time- 1.5 Hours			Max.Marks- 50
Instructions - 1) Answer all questions. 2) Assume suitable data.			

Q.1	Solve any four	20 marks
	a. Explain how the Domain Naming System (DNS) works.	5 marks
	b. How vulnerability is different from threat explain with an example.	5 marks
	c. List out the different information received in the *response* method of the *HTTP* protocol.	5 marks
	d. Explain server-side request forgery with an example.	5 marks
	e. Write a short note on *HTTP* fingerprinting	5 marks
Q.2	Attempt all	15 marks
	a. Explain the Vulnerability management lifecycle.	5 marks
	b. How does an attacker take advantage of security misconfiguration?	5 marks
	c. Write a short note on a cookie. How does it help the attackers?	5 marks
Q.3	Attempt a and b	15 marks
Q.3 a	Attempt any one	
Q.3 a	I. Explain the *Common Vulnerability Scoring System* in detail.	8 marks
	OR	
	II How do you assess the vulnerability of a Fintech company? Explain the different steps involved in this.	8 marks
Q.3 b	Attempt any one	7 marks
Q3 b	I. Explain the Secure Source Code Review process, why it is required, and under what circumstances you will prefer to go for automation.	7 marks
	OR	
	II. Explain *STRIDE-based* threat modeling and how it differs from the *DREAD* model.	7 marks

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NATIONAL FORENSIC SCIENCE UNIVERSITY

Branch: MSc Cyber Security/ MSc DFIS **Sem:** I **Date:** 14/10/2024

Subject name: Cyber security audit and compliance

Subject code: CTMSCS SI P2/CTMSDFIS SI P2

Time: 1.5 hours **Max Marks:** 50

Instructions: 1) Answer all the questions. 2) Assume suitable data.

Q1.	Solve any Four	20 marks
	I) How do Security assessment & security audit differ from each other?	5 marks
	II) Why are governance & compliance important?	5 marks
	III) What does an organisation do to be in compliance?	5 marks
	IV) What if the organisation does not comply with compliance laws?	5 marks
	V) What are we actually auditing within the IT infrastructure?	5 marks
Q2	Attempt All	15 marks
	I) What are the key aspects of Cybersecurity Audit?	5 marks
	II) List examples of security controls in IT Audit. Firewall, [illegible], IP, logs	5 marks
	III) Explain internal-audit vs external-audit	5 marks
Q3	Attempt a and b	15 marks
Q3 a.	Attempt any one	8 marks
	I) Why are security controls important in an IT audit?	
	OR	
	II) Explain Goal-based controls & Implementation based controls with examples.	
Q3 b.	Attempt any one	7 marks
	I) Explain seven domains of typical IT infrastructure	
	OR	
	II) Discuss the steps involved in planning & implementation of IT Audit for compliance	

→ Risk mgmt
→ IR P review
→ Security flaws identify
→

AD

Seat No.: ________ Enrolment No. 24014745 2024

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (December – 2024)

M.Sc. Cyber Security, Semester – I

Subject Code: CTMSCS S1 P2 **Date: 05/12/2024**

Subject Name: Cyber Security Audit and Compliance

Time: 02:30 PM to 05:30 PM **Total Marks: 100**

Instructions:

1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

			Marks
Q.1		**Attempt any three.**	
	(a)	Why are governance and compliance important.	08
	(b)	Differentiate between Security Assessment and Security Audit.	08
	(c)	Define IT security audit and explain its objectives.	08
	(d)	Explain IT security Assessment, IT Security Audit and Security compliance.	08
Q.2		**Attempt any three.**	
	(a)	What is meant by scope of an audit? Explain its importance in the auditing process.	08
	(b)	How important is an IT Audit process.	08
	(c)	What is Computer Assisted Audit Techniques (CAATs), and why are they important.	08
	(d)	Write difference between Audit plan and Audit process.	08
Q.3		**Attempt any three.**	
	(a)	How to identify the risk level and how to write IT infrastructure audit report.	08
	(b)	What are the seven domains of a typical IT infrastructure? Briefly describe each.	08
	(c)	What are the key components of an IT infrastructure audit report?	08
	(d)	How to maximize CIA with the help of Audit and compliance.?	08
Q.4		**Attempt any two.**	
	(a)	What is risk analysis, and why is it important for organizations?	07
	(b)	What are the key phases of the BCP life cycle?	07
	(c)	How does disaster recovery planning differ from business continuity planning?	07
Q.5		**Attempt any two.**	
	(a)	Briefly describe the concept of Cyber Terrorism defined under the IT Act.	07
	(b)	Explain the purpose of ISO/IEC 27002 and its role in supporting ISO/IEC 27001.	07
	(c)	Discuss the significance of HIPAA in protecting healthcare information.	07

— End of Paper—

National Forensics Sciences University, Goa Campus
TA-1 Examination

Program Name – M.Sc Cyber Security	Sem – 1	Date- 10/09/2024
Subject Name- Cyber Security Audit & Compliance		
Subject Code- CTMSCS SI P2		
Time- 45 minutes		Max. Marks- 25
Instructions - 1) Answer all questions. 2) Assume suitable data.		

	Answer any 5 questions (5x5 marks each)	25 marks
Q1	Explain Types of IT audit in detail.	5 Marks
Q 2	Describe how to conduct IT Audit and process of IT security audit.	5 marks
Q3	Compare Internal audit vs External Audit.	5 marks
Q4	List Best Practices to Perform Cyber Security Audits.	5 marks
Q5	What is compliance and describe the importance of compliances.	5 marks
Q6	Why are assessments and audits important?	5 marks

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National Forensics Sciences University, Goa Campus
TA-1 Examination

Program Name – BSc-MSc Forensic science (5-year integrated) Sem – I
Date- 11/09/2024
Subject Name- Introduction to Forensic Science and Cyber Law
Subject Code- CTMS_CS/DFIS/AIDS SI P5
Time- 45 minutes Max. Marks- 25
Instructions - 1) Answer all questions. 2) Assume suitable data.

Q.1	Multiple Choice Questions (1 mark each)	10 marks
	I. Find the correct order of forensic science laboratories set up in India- a. CFSL>SFSL>RFSL>MFSL b. MFSL>RFSL>SFSL>CFSL c. SFSL>CFSL>MFSL>RFSL d. RFSL>SFSL>CFSL>MFSL	1 mark
	II. The first fingerprint bureau was established in a. Delhi b. Chennai c. Kolkata d. Shimla	1 mark
	III. The first central forensic Science laboratory (CFSL) in India was established in a. Kolkata b. Delhi c. Hyderabad d. Chandigarh	1 mark
	IV. "When two objects come in contact there is always an exchange of materials". This statement was given by ________________.	1 mark
	V. Law of comparison, compares between a. People with dissimilar likes b. People from extended family c. Identical twins d. People with similar likes	1 mark
	VI. The role of the Forensic Document Examiner is to- a. Verify the documents involved in forgery case b. Verify the handwriting c. Verify the Ink used d. All of the above	1 mark
	VII. Study of human skeletal structure comes under a. Forensic Chemistry b. Forensic Anthropology c. Forensic Pathology a. Forensic Entomology	1 mark

	VIII.	Identification of volatile and non-volatile poisons of organic and Inorganic compounds carried out in ________ a. Biology division b. Chemistry division c. Toxicology division d. Physics division	1 mark
	IX.	Choose the correct role of DFSS a. Promote research and development in forensic field b. Guide, regulates and controls the working of FSLs c. It provides scientific aid and criminal justice systems d. All	1 mark
	X.	"The prompt action is necessary in all aspects of criminal investigation". This sentence is best suited for which principle- a. Law of analysis b. Law of Progressive change c. Law of circumstantial fact d. Law of Probability	1 mark
Q2	Answer any 3 questions (3x5 marks each)		15 Marks
	i.	Explain, "There is an urgent need for Forensic experts in India".	5 marks
	ii.	Discuss the various laws of Forensic Science.	5 marks
	iii.	Why Forensic science is a multi-disciplinary field? And what is its significance?	5 marks
	iv.	State the organization order of various forensic science laboratories in India and their function.	5 marks

Enrolment No. 119

NATIONAL FORENSIC SCIENCES UNIVERSITY
School of Cyber Security, M.Sc. Cyber Security Sem I
Artificial Intelligence
Mid Semester Examination Semester – I – Oct.-2025

Subject Code: CTMSCS SI P4 Date:10/10/2025
Subject Name: Artificial Intelligence
Time: 04:00PM TO 05:30PM Total Marks: 50

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.

Q.1.	Answer the following / Define the following	10*2 (Marks)-20
	1. A team is developing a neural network to classify support tickets into categories. Describe the purpose of the Compile-Train-Validate cycle in this workflow. 2. A data analyst is summarizing customer age data. In what situations would they prefer Mean, Median, or Mode? 3. A teacher uses past student performance data to predict future scores. Identify the type of learning involved and illustrate with a suitable example. 4. A researcher is analyzing the consistency of test scores across different batches. How would Standard Deviation and Variance help in this analysis? 5. A machine learning model performs well on training data but poorly on new data. Explain this issue using the concepts of Overfitting and Underfitting. 6. A developer is building a binary classifier using a single-layer neural network. Describe the role of a Perceptron in this setup. 7. A marketing team wants to understand the relationship between advertisement spend and sales. How can Classification and Regression be used to support their decision-making? 8. A data scientist is evaluating multiple models for predicting customer churn. Each model shows different Loss values during training. Explain the role of a Loss Function in this context, and how it influences model optimization and performance 9. Describe Convolutional Neural Network (CNN) architecture MNIST data. 10. A neural network fails to learn complex patterns. How does the choice of Activation Function influence the learning capability of the model?	

Enrolment No. ____________

Q.2.	Answer the following question in short (any 5)	5*4 (Marks)-20
	1. A startup is building a recommendation engine for its e-commerce platform. Based on the nature of available data and desired outcomes, how would you decide which type of Machine Learning–Supervised, Unsupervised, or Reinforcement–is most appropriate? Support your answer with examples. 2. A data scientist splits a dataset into three parts before training a model. Explain how the Training, Validation, and Testing sets are used in this workflow and why each is critical for building a reliable model. 3. While evaluating two models for predicting loan defaults, one shows higher Accuracy but also higher Loss. How would you interpret these metrics, and which model might be more trustworthy in deployment? 4. A team is tuning a machine learning model to improve performance. They adjust learning rate, batch size, and number of layers. Explain what Hyperparameters are, and how they differ from model parameters in this context. 5. A company wants to segment its customer base without labelled data. Explain how Unsupervised Learning can be applied here, and mention two models that could be used effectively. 6. A researcher wants to reduce model complexity and improve interpretability by selecting only the most relevant features. Describe the process of Feature Selection and its impact on model performance.	
Q.3.	Answer the following in detail	1*10 (Marks)-10
	1. Explain the architecture of an Artificial Neural Network (ANN). Describe in detail the process of training a multi-layer ANN model, including backpropagation and forward pass. 2. Explain the complete workflow of building and evaluating a Convolutional Neural Network (CNN) model. Include detailed explanations of convolutions.	
Total		50 Marks

National Forensics Sciences University, Goa Campus
TA-1 Examination

Program Name – M.Sc. CS	Sem – I	Date- 10.02.25
Subject Name- Incident Response & Digital Forensics	Subject Code- CTMSCS SII P4	
Time- 45 minutes		Max. Marks- 25
Instructions - 1) Answer all questions. 2) Assume suitable data.		

Q.1	Multiple Choice Questions (1 mark each)	10 marks
	1a. What is the primary goal of developing an incident response plan in web security? a. Improving website aesthetics b. Actively blocking all incoming and outgoing traffic c. Minimizing the impact of security incidents and ensuring a structured response d. Granting unrestricted access to all users	1 mark
	1b. What role does "stakeholder communication" play in the development of an incident response plan? a. Improving website aesthetics b. Enhancing server performance c. Ensuring clear communication channels and responsibilities during incidents d. Granting unrestricted access to all users	1 mark
	1c. Why is it important to conduct regular testing and drills of an incident response plan? a. Improving website aesthetics b. Ensuring the plan is up-to-date and the team is prepared for real incidents c. Enhancing server performance d. Granting unrestricted access to all users	1 mark
	1d. A loss of ____ is the unauthorized disclosure of information. a. Confidentiality b. Authenticity c. Integrity d. Availability	1 mark

	1e. Which of the following usually observes each activity on the internet of the victim, gathers all information in the background, and sends it to someone else? a. Malware b. Spyware c. Adware d. None of the above	1 mark
	1f. In the context of incident identification, what is the role of security information and event management (SIEM) tools? a. Improving website aesthetics b. Actively blocking all incoming and outgoing traffic c. Collecting and analyzing security event data for signs of incidents d. Granting unrestricted access to all users	1 mark
	1g. What describes the immediate action taken to isolate a system in the event of a breach? a. Containment b. Eradication c. Recovery d. None	1 mark
	1h. Is the following statement true or false? 'Incident response is a structured methodology for handling security incidents, breaches, and cyber threats.' a. True b. False	1 mark
	1i. Under which plan does personnel perform business processes in an alternate manner until normal operations resume? a. Disaster recovery plan (DRP) b. Business continuity plan (BCP) c. Business impact analysis (BIA) d. None	1 mark
	1j. In the context of incident identification, what role does user awareness play? a. Improving website aesthetics b. Actively blocking all incoming and outgoing traffic c. Encouraging users to report suspicious activities for early identification d. Granting unrestricted access to all users	1 mark
Q.2	Answer any 3 questions (3x5 marks each)	15 Marks
	i. Explain the different stages of incident response procedure.	5 marks

	ii.	How do you categorize different types of incidents?	5 marks
	iii.	How do you detect/identify incidents in your organization, explain.	5 marks
	iv.	Explain the proactive and reactive incident detection.	5 marks

Enrolment No. 24034700 2004

NATIONAL FORENSIC SCIENCES UNIVERSITY

School of Cyber Security & Digital Forensics

End Semester Examination - May-2025

Subject Name: Malware Analysis **Total Marks: 50**

Use Volatility to list processes from live memory.

Define kernel debugging and its differences from user-mode debugging.

Nationa... nsics Sciences University, Goa Campus

TA-1 Examination

Program Name – M.Sc. Cyber Security Date- 09/09/25 Subject Name- Artificial Intelligence Time- 45 minutes	Sem – I Subject Code- (CTMSCS SI P4) Max. Marks- 25

I	**Short Answer Questions**		**10 marks**
	I.	Differentiate between a Python module and a package, detailing their creation and import mechanisms. Additionally, explain the role and functionality of context processors in Python-based web frameworks.	2 marks
	II.	Design a Python class called DataProcessor that encapsulates statistical operations. The class should have methods to calculate mean, median, and standard deviation of a dataset. Include a class method to create an instance from a list and demonstrate its usage with sample data.	2 marks
	III.	Develop a Python snippet that constructs a dictionary mapping student name to their scores, ensuring typecasting from strings to integers and validating that each score falls within an acceptable range of 0-100. Include at least three entries.	2 marks
	IV.	What is a data class in Python? Mention one benefit of using data class over a regular class with example.	2 marks
	V.	List two key differences between NumPy arrays and Python lists. Focus on performance and functionality.	2 marks
II	**Answer any 3 questions (3 x 5 marks each)**		**15 Marks**
	i.	Explain how to set up a Python development environment using Anaconda, including creating a new environment, installing NumPy and Pandas, and computing the cube of the matrix [1, 0, -1; 1, 0, 0; 1, -1, 2].	5 marks
	ii.	the role of loops and conditional statements in Python programming. Write a script that takes a list of integers and prints whether each number is even or odd, using both loop and condition, also time the script using decorators.	5 marks
	iii.	Describe how Matplotlib and SciPy can be used together for scientific visualization and computation. Provide an example where you plot a sine wave using Matplotlib and compute its sum of series over interval [π/12, π/3].	5 marks
	iv.	List all bugs in following program: import maths def compute_circle_area(radius) if radius <= 0:	5 marks

National Forensic Sciences University
School of Cyber Security & Digital Forensics
M.Sc. CS (2024-25)

Subject Code: (CTMSCS SII P3) **Date: 10/02/2025**
Subject Name: Malware Analysis
Exam: TA- I Examination (Feb 2025) **Time: 03:00 PM to 03:45 PM**

Q1. Answer the following questions in brief. (Answer any 3) [15 Marks]

1. Describe the different types of malware and methods of malware analysis, and explain how Clam AV virus signatures are utilized for malware detection.

2. Describe the working algorithm FLOSS tool with one use case.

3. Describe the X86 CPU architecture, including the processing unit (CPU) and memory (register and stack).

4. Translate a simple C code snippet to assembly and explain the underlying constructs.

5. Discuss the YARA signatures, providing an example of rule formation.

Q2. Answer the following questions in word(s). (Attempt all) [10 Marks]

1. ________ type of malware is known for disguising itself as a legitimate program.

2. List two tools primarily used to set up a malware analysis laboratory.

3. Hashing is used in malware analysis to ________.

4. ________ section of a PE file contains the executable code.

5. ________ is the primary function of Clam AV Virus Signature.

6. ________ tool is used for capturing and analyzing network traffic in malware analysis.

7. List two components present in X86 CPU architecture.

8. List any two tools used for debugging and analyzing executable files.

9. List any two tools used for creating virus signatures and give one example showing its use case.

10. List one tool used for live memory analysis and give one example showing its use case.

Enrolment No.________________________

NATIONAL FORENSIC SCIENCES UNIVERSITY

School of Cyber Security & Digital Forensics

M.Sc. CS Sem - II

Mid Semester Examination - March-2025

Subject Code: (CTMSCS SII P3) **Date: 18/03/2025**

Subject Name: Malware Analysis

Time: 11:00 AM to 12:30 AM **Total Marks: 50**

Instructions:

1. **Make suitable assumptions wherever necessary.**
2. **Figures to the right indicate full marks.**

Q.1 Answer the following / Define the following **10 x 2 (Marks)=20**

i) What is the forensic importance of malware analysis?

ii) Mention one behavior commonly observed in malware.

iii) What is the significance of a virtual machine in a malware analysis lab?

iv) List two reverse engineering tools used for malware analysis.

v) Define the role of PE (Portable Executable) file headers.

vi) What is the purpose of a malware signature?

vii) What is the role of Process Monitor?

viii) Mention one use of Wireshark.

ix) What is IDA Pro used for?

x) Translate a basic C code example into assembly (simplified representation).

Q.2 Answer the following questions in short (any ~~5~~ 4) 5 **4 x 5 (Marks)=20**

i) Describe the role of Dependency Walker in malware analysis.

ii) What are YARA signatures, and how are they used in malware detection?

iii) Explain the usage of Process Explorer in identifying malware behavior.

iv) Differentiate between kernel mode and user mode debugging.

v) Discuss the significance of packers in malware and the process of unpacking them.

vi) Discuss anti-disassembly techniques employed by malware.

Q.3 Answer the following in detail (any 2) **2 x 5 (Marks)=10**

i) Explain how sandboxes are utilized to safely monitor malware behavior.

ii) Explain debugging features such as breakpoints, exceptions, and program execution modification in detail.

iii) Analyze the differences between anti-debugging and anti-virtual machine techniques.

National Forensics Sciences University, Goa Campus
Mid Semester Examination

Branch – M.Sc. Cyber Security /M.Tech. A.I. & D.S. **Sem** – II
Date- 19/03/2025
Subject Name - Mobile Security /Mobile Security and Forensics
Subject Code - CTMSCS SII P3 / CTMTAICS SII P2
Time- 1.5 Hours **Marks- 50**
Instructions - 1) Answer all questions. 2) Assume suitable data.

Q.1	Solve any four	20 marks
	a. What is Application Permission? Explain different types of Application permission protection level	5 marks
	b. What is application sandboxing? Why it is important for mobile security?	5 marks
	c. What is Pen-testing? Explain different strategies for Pen-Testing.	5 marks
	d. What is Secure Inter Process Communication?	5 marks
	e. Explain Android File systems in detail.	5 marks
Q.2	Attempt all	15 marks
	a. Discuss Android Boot Process. Explain the steps involved in the boot process. ~~the binder framework~~.	5 marks
	b. Use a pictorial representation to describe the android architecture.	5 marks
	c. What is ADB Command? Explain 5 ADB Commands in detail.	5 marks
Q.3	Attempt a and b	15 marks
Q.3 a	Attempt any one	
Q.3 a	I) Explain the OWASP top 10 vulnerabilities for Mobiles?	8 marks
	OR	
	II)What is Hexdump? Explain the important features of Hexdump.	8 marks
Q.3 b	Attempt any one	7 marks
Q3 b	i) Explain the difference between DALVIk and SMALI.	
	OR	
	ii) What is reverse engineering? What are the steps involved in reverse engineering of a software	7 marks

Seat No.: ________ Enrolment No.__________

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (APRIL – 2025)

M.Sc. Cyber Security- Semester - II

Subject Code: CTMSCS SII P1 **Date: 24/04/2025**

Subject Name: Network Security

Time: 10:30 AM to 01:30 PM **Total Marks: 100**

Instructions:

1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Use of Scientific Calculator is allowed.**

			Marks
Q.1		**Attempt any three.**	
	(a)	Compare the ISO/OSI model with the TCP/IP model. Highlight the similarities and differences between the layers of both models and describe how each OSI layer maps to the respective layers in the TCP/IP model.	08
	(b)	What is the CIA triad in network security? Describe each of its components and support your explanation with a practical example for Confidentiality, Integrity, and Availability.	08
	(c)	What is MAC flooding? Discuss the sequence of actions an attacker takes to perform a MAC flooding attack on a network switch.	08
	(d)	An organization wants to secure its network from internal and external threats. Explain how implementing IDS and IPS can help achieve this goal.	08
Q.2		**Attempt any three.**	
	(a)	Outline the complete process of a penetration test, from planning and scoping to reporting the findings.	08
	(b)	Describe the working mechanism of any symmetric encryption algorithm in detail.	08
	(c)	Assume you are setting up a basic RSA encryption system using the primes p=19 and q=23, and a public exponent e=13. Complete the following tasks: i) Determine the public and private keys. ii) Encrypt the message M=3 using RSA encryption.	08
	(d)	Explain the difference between stream cipher and block cipher in detail.	08
Q.3		**Attempt any three.**	
	(a)	Explain the concept of a Digital Signature, and provide a detailed explanation of how the MD5 hash function works.	08
	(b)	Define the terms "attack surface" and "attack vector." Explain each in detail, highlighting their significance in cybersecurity. Additionally, discuss common types of attacks associated with each and suggest countermeasures.	08
	(c)	Explain the methods used by war driving and dumpster diving to collect information. How can businesses protect themselves from these tactics?	08
	(d)	Explain OSCAR Methodology with examples.	08

3648

61

Q.4		**Attempt any two.**	
	(a)	Provide a detailed explanation of the Transport Layer Security (TLS) protocol, its purpose, and how it works. Also, explain the role of SSL in HTTPS and how it ensures a secure connection between a client and server.	07
	(b)	Explain 802.11 protocol and also discuss the differences between WEP and WPA.	07
	(c)	What is the role of an Access Point (AP) in a WLAN? How does it facilitate communication between wireless and wired devices?	07
Q.5		**Attempt any two.**	
	(a)	What is live forensics, and how does it differ from traditional digital forensics? Explain the key steps involved in conducting a live forensic investigation, and provide examples of the types of evidence that can be gathered in a live forensics scenario.	07
	(b)	Imagine a business suspects unauthorized access to its critical systems. As a digital forensic investigator, explain the steps you would follow in a network forensic investigation. i. What network-based digital evidence would you focus on first? ii. Describe the process of gathering evidence in a network forensic investigation. iii. What challenges do you face in ensuring the admissibility and integrity of digital evidence?	07
	(c)	Define the following terms and explain their roles in network management (Any three): i. DNS Server ii. DHCP Server iii. Proxy Server iv. SOC	07

--- End of Paper---

Seat No.: ________ Enrolment No.____________

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (April – 2025)
M.Sc. Cyber Security Semester – II

Subject Code: CTMSCS SII P2 **Date: 28/04/2025**
Subject Name: Malware Analysis
Time: 10.30 hrs. – 13.30 hrs. **Total Marks: 100**

Instructions:
1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

			Marks
Q.1		**Attempt any three.**	
	(a)	Discuss process explorer and process monitor tools.	08
	(b)	i) What is unpacking stub? ii) What is keylogger? How it functions?	08
	(c)	Explain any two encoding techniques with example.	08
	(d)	Convert the following disassembly into the C construct code. Calculate the last value of eax, ebx, ecx and edx registers from following disassembly.	08

```
00401006    mov [ebp+var_4], 0Ah
0040100D    mov [ebp+var_8], 9
00401014    mov eax, [ebp+var_4]
00401017    add eax, 0Bh
0040101A    mov [ebp+var_4], eax
0040101D    mov ecx, [ebp+var_4]
00401020    sub ecx, [ebp+var_8]
00401023    mov [ebp+var_4], ecx
00401026    mov edx, [ebp+var_4]
00401029    sub edx, 1
0040102C    mov [ebp+var_4], edx
0040102F    mov eax, [ebp+var_8]
00401032    add eax, 1
00401035    mov [ebp+var_8], eax
00401038    mov eax, [ebp+var_4]
0040103B    cdq
0040103C    mov ecx, 3
00401041    div ecx
00401043    mov [ebp+var_8], edx
```

Q.2		**Attempt any three.**	
	(a)	A malware is using a technique which injects code into another running process, and that process executes the malicious code. What kind of technique is this? Explain that in detail with example.	08
	(b)	Write a C program for calculating the simple interest, create its assembly code and explain.	08
	(c)	Write a detailed note on types of malwares with its examples.	08

(d) Write a detailed and real-world case study of malware attack with all the technical information. 08

Q.3 **Attempt any three.**

(a) Discuss any 08 windows functions/APIs with its malicious use case. 08

(b) Explain the basic static analysis process with a complete example. 08

(c) Malware author uses some techniques to deviate the analyst and makes its analysis difficult for them. Which technique(s) the malware author use here to perform the said task? Explain that/those in detail. 08

(d) Discuss the process of reverse engineering of APK file, also explain the possible suspicious artefacts from its reverse engineering. 08

Q.4 **Attempt any two.**

(a) Write a detailed note on Ollydbg and its features for advance dynamic analysis. 07

(b) Write a short note on anti-malware signature and explain the practical of clam-av to generate the anti-malware signature. 07

(c) Write a detailed note on IDA Pro tool with its functions for advance static analysis. 07

Q.5 **Attempt any two.**

(a) What is the memory of CPU except cache, known as? Explain that memory in detail. 07

(b) Explain the fake net and its significance. How the fake-net can be created practically? 07

(c) Discuss the android architecture in detail. 07

— End of Paper---

Seat No.: ________ Enrolment No. ________

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (April – 2025)
M.Sc. Cyber Security (Batch 2024-2026)
Semester – II

Subject Code: CTMSCS SII P3
Subject Name: Mobile Security
Time: 10:30 AM to 01:30 PM

Date: 25/04/2025
Total Marks: 100

Instructions:

1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

			Marks
Q.1		**Answer the following question (Attempt any three)**	
	(a)	Describe Android Architecture in detail with an appropriate diagram.	08
	(b)	Describe Android Application Components in detail.	08
	(c)	What is Sandboxing? Describe Inter-process Communication in the Android OS.	08
	(d)	Explain the Android Boot process in detail.	08
Q.2		**Answer the following question (Attempt any three)**	
	(a)	What is ADB? Explain any five ADB commands.	08
	(b)	Discuss access control issues and how you would pen-test it.	08
	(c)	What is Content Provide Leakage, and how do you pent-test it?	08
	(d)	Discuss the client-side injection with an example.	08
Q.3		**Answer the following question (Attempt any three.)**	
	(a)	Discuss the Mobile application security pen-testing strategy.	08
	(b)	Discuss any four Drozer modules with examples.	08
	(c)	Discuss the Static Analysis using MobSF.	08
	(d)	Discuss the insecure data storage issue with an example.	08
Q.4		**Answer the following question (Attempt any two)**	
	(a)	Discuss how Frida and the Objection Framework will be used to pen-test mobile application security.	07
	(b)	Discuss reverse engineering techniques to reverse engineer an Android application.	07
	(c)	Discuss the Dynamic Analysis using MobSF.	07
Q.5		**Answer the following question (Attempt any two)**	
	(a)	What is the importance of Network Traffic Analysis of an Android Device? How does it help us in penetration testing? Explain.	07
	(b)	Describe the difference between Active and Passive network traffic analysis of an Android Device in detail.	07
	(c)	Describe the Android Traffic Interception. How can you intercept the HTTP/HTTPS traffic using a Proxy Server? Explain in detail.	07

--- End of Paper---

Seat No.: ________ Enrolment No.__________

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination

M. Sc. Cyber Security - Semester – II – April - 2025

Subject Code: CTMSCS SII P4 **Date:** 29/04/2025

Subject Name: Incident Response & Digital Forensics

Time: 10:30 AM to 01:30 PM **Total Marks:** 100

Instructions:

1. **Write down each question on separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

			Marks
Q.1		**Answer the following (Attempt Any Three)**	
	(a)	i) Define Incident Response & its need in the domain of cyber security. ii) What are the goals of Incident Response?	08
	(b)	Your company has recently hired few interns to help you out in your IRM team. Give a brief idea to the team about the signs of an incident with proper examples.	08
	(c)	Explain the categories of Incident Response.	08
	(d)	Your department head has called an urgent meeting to discuss the security of client information and you being a security consultant he has asked you to explain the different data classification techniques to the team.	08
Q.2		**Answer the following (Attempt Any Three)**	
	(a)	As the cybersecurity lead for a healthcare facility undergoing a ransomware attack that compromises patient records, outline your strategy for incident prioritization based on informational impact. Additionally, provide an example for each category of impact.	08
	(b)	How virtualization affects the phases of incident handling?	08
	(c)	How can an organization know how much would an incident cost to them? Explain with an example.	08
	(d)	Describe the various roles within an incident response team along with their primary responsibilities.	08
Q.3		**Answer the following (Attempt Any Three)**	
	(a)	What are the main sources of logs used in Incident Response Management? Briefly describe at least three different log sources and explain how each one aids in understanding and addressing security incidents within an organization's network.	08
	(b)	Explain preparation, identification and eradication phases of Incident handling.	08
	(c)	Prepare an Incident handler checklist taking a case scenario.	08
	(d)	Discuss the phases of containment, eradication & recovery in context to Incident Response Management.	08

Q.4 **Answer the following (Attempt Any Two)**

(a) Explain Digital Forensics and its branches. Also state the importance of Locard's Principle of Exchange in Digital Forensics. **07**

(b) What is Chain of Custody and explain its necessity in Digital Forensics? Also state how is integrity maintained in Digital Forensics. **07**

(c) Answer the following : **07**

i) Write Blockers

ii) Imaging v/s Cloning

Q.5 **Answer the following (Attempt Any Two)**

(a) Explain NTFS & FAT32 File Systems. **07**

(b) What are Registry Artefacts and how are they important in establishing a case in Digital Forensics? Explain with examples and artefacts. **07**

(c) Consider the following Sysmon Event Log Entry and answer the asked question: **07**


```
{
 "Event": {
  "Timestamp": "2023-05-20T13:45:55.000Z",
  "EventID": 1,
  "Provider": "Microsoft-Windows-Sysmon",
  "Channel": "Microsoft-Windows-Sysmon/Operational",
  "Computer": "SERVER-EXAMPLE",
  "EventData": {
   "Image": "C:\\Windows\\System32\\rundll32.exe",
   "ParentImage": "C:\\Windows\\System32\\svchost.exe",
   "ParentCommandLine": "svchost.exe -k netsvcs",
   "TargetImage": "C:\\ProgramData\\Malicious\\malware.exe",
   "TargetCommandLine": "malware.exe -start -mode stealth",
   "GrantedAccess": "0x1410",
   "LogonId": "0x3e7",
   "UtcTime": "2023-05-20 13:45:54.500"
  }
 }
}
```


Explain the following event log parameters & their significance and also Identify the potential threat through the provided event log.

--- End of Paper---

National Forensics Sciences University, Goa Campus

M.Sc. CS and MSc DFIS - Semester -II

Mid- Semester Examination

Subject Code: CTMSCS SII P1/ CTMSDFIS SII P3 — **Date: 18/03/2025**

Subject Name: Network Security and Forensics

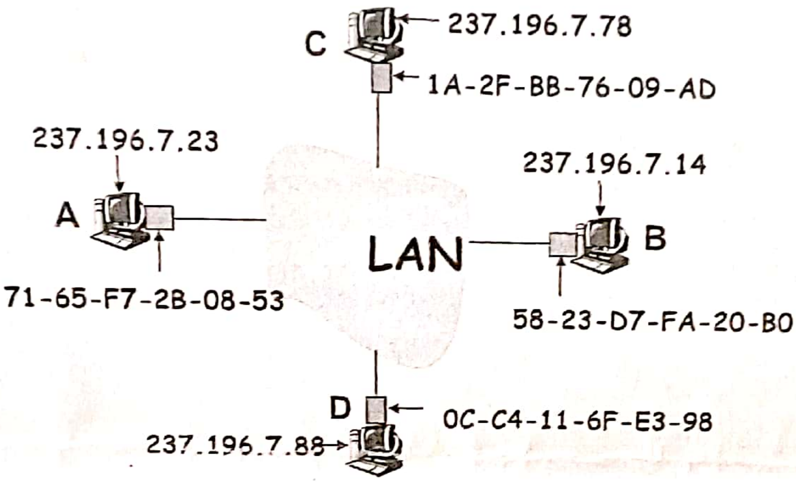
Time: 90 Minutes — **Total Marks: 50**

Instructions –

1) Answer all questions.
2) Assume suitable data.
3) Scientific Calculator is allowed.
4) Parts of the question should attend the same place.

Q.1 Attempt all. — **20 marks**

(a) — 5 Marks



Consider the above network topology, **User A** wants to communicate with **User B**. Explain the explain ARP protocol with respect to this scenario. Further consider **User C** as the attacker and explain the ARP spoofing in the same topology. Also highlight the all-possible attack vectors and attack surfaces.

(b)	A company wants to set up a web server accessible from the internet. Explain the role of the DNS server in this scenario.	5 Marks
(c)	Encrypt the following message using **Playfair cipher**. Message: **jacuzzi operation** Key: **extrajudicially**	5 Marks
(d)	(i) Calculate the power modulo, 191^{930} ***mod 103.*** (ii) What is the purpose of TTL field in IP packet? How is TTL useful for forensic purpose?	5 Marks
Q.2	Attempt three.	**15 Marks**
(a)	Write note on DOS and distributed denial-of-service (DDoS) attacks.	5 marks
(b)	What would you do if ***nmap*** port scans are blocked by network security administrator? How would you gather host information in such case?	5 marks
(c)	Explain the differences between error control and flow control.	5 marks
(d)	Explain the WLAN architecture?	5 marks

Q.3	Attempt all.	15 Marks
(a)	Use two global prime number ***17*** and ***31***, the value of ***e*** is ***7*** and message ***M*= 3**, calculate the public key, private key, and the corresponding cipher text. Also prove that *RSA* decryption is the inverse of *RSA* encryption.	08 Marks
	OR	
	Use two global prime number ***37*** and ***43***, the value of ***e*** is ***71*** and message ***M*= 2**, calculate the *public key*, *private key*, and the corresponding cipher text. Also prove that *RSA* decryption is the inverse of *RSA* encryption.	
(b)	*Alice* and *Bob* wish to swap keys by using *Diffie-Hellman* key exchange algorithm and are agreed on prime ***p* = 23** and base or generator is ***g*= 5.** Calculate the *secret key* of each user and *shared session key* for both the users. Also explain with the same question that how can *Eve* (untrusted third person) exploit *Man-in-Middle attack*.	07 Marks
	OR	
	Alice and *Bob* wish to swap keys by using *Diffie-Hellman* key exchange algorithm and are agreed on prime ***p* = *31*** and base or generator is ***g*= *3***. Calculate the secret key of each user and shared session key for both the users. Also explain with the same question that how can *Eve* (untrusted third person) exploit Man-in-Middle attack.	

~~~~END OF PAPER~~~~

e	x	t	r	a
u	d	i	c	l
y	b	f	g	h
k	M	N	o	P
q	s	v	w	z

1.25 mod

Seat No.: ________ Enrolment No. ________

NATIONAL FORENSIC SCIENCES UNIVERSITY

PRACTICAL EXAMINATION MAY 2025

Date: 05/05/2025

Subject Code: CTMSCS SII L1/ CTMSDFIS SII L1 **Total Marks: 50**

Subject Name: Network Security Lab/ NSF LAB **Time: 30 Minutes**

Instructions:

1. **Attempt all questions.**
2. **Figures to the right indicate full marks.**
3. **Assume Suitable data.**

SET-1: Roll Number ending with 4, 8, 12, 16, 20, 24 28
SET-2: Roll Number ending with 3, 7, 11, 15, 19, 23 27
SET-3: Roll Number ending with 2, 6, 10, 14, 18, 22 26
SET-4: Roll Number ending with 1, 5, 9, 13, 17, 21 25

SET		Questions	Marks
SET-1	Q.1	Write a Python program that implements the ***RSA encryption algorithm***. Demonstrate its usage by encrypting, decrypting and digital signature a given message.	
	Q.2	Imagine a scenario where a distributed denial of service (DDoS) attack is targeting your organization's web servers. Outline the steps you would take to mitigate the impact and prevent future occurrences.	
SET-2	Q.1	Configure Quality of Service parameters in ***Cisco Packet Tracer*** to prioritize specific types of network traffic of NFSU website. Test and verify the effectiveness of QoS settings in managing network bandwidth.	
	Q.2	In the event of a security incident, such as a data breach or malware infection, describe the steps you would take to contain the incident, investigate the root cause, and communicate effectively with stakeholders.	25
SET-3	Q.1	Requirement (an open-source rainbow table-based password cracking tool) Hash values of passwords to crackOptional: Any additional rainbow tables or wordlists Task: • Compare the cracked passwords with the original list. • Note how many passwords were successfully cracked.	
	Q.2	An employee's laptop has been lost or stolen. Describe the steps you would take to ensure that sensitive data on the device remains secure and cannot be accessed by unauthorized individuals.	
SET-4	Q.1	Task: Perform a vulnerability assessment on a NFSU website to identify open ports, running services, and potential weaknesses. Compile a report with identified vulnerabilities and remediation suggestions. Tools: Nmap, Nessus, OpenVAS, Metasploit, etc. Objectives: Discover network vulnerabilities, evaluate risk levels, and propose security improvements to harden the network.	
	Q.2	Imagine an employee falls victim to a phishing attack and unknowingly installs malware on their workstation. Detail the measures you would take to detect and remove the malware, as well as prevent similar incidents in the future.	

END OF PAPER

Enrolment No.___________

NATIONAL FORENSIC SCIENCES UNIVERSITY
GOA CAMPUS

M.Sc. DFIS - Semester -II/ M.Sc. Cyber Security Term Assessment-I

Subject Code: CTMSDFIS SII P1/ CTMTCS SII P1 **Date: 11/02/2025**
Subject Name: Network Security & Forensics/ Network Security
Time: 45 Minutes
Instructions: **Total Marks: 25**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

Q1 To Q10 each for 1 mark (10x1=10)

Fill the appropriate answer:

Q 1 The process of adding headers and trailers to data when it moves down the OSI layers is called_____

Q 2 A router uses a___________ to determine the path for forwarding packets.

Q 3 A ___________. can connect two networks with different architectures, protocols, or data formats

Q 4 The_____________ layer of the OSI model is responsible for end-to-end communication.

Q 5 In the Playfair cipher, if two letters in a digraph appear in the same row of the matrix, they are replaced by the letters immediately __________ in the same row.

Q 6 The __________ attack is a cryptanalysis method that attempts to deduce the encryption key by analyzing multiple ciphertexts encrypted with the same key.

Q 7 The Vigenère cipher is a type of __________ cipher that uses a repeated keyword to encrypt plaintext.

Q 8 A cryptographic system is considered __________ secure if the ciphertext provides no additional information about the plaintext beyond what is already known.

Q 9 A product cipher is a combination of __________ and __________ techniques applied repeatedly to enhance security.

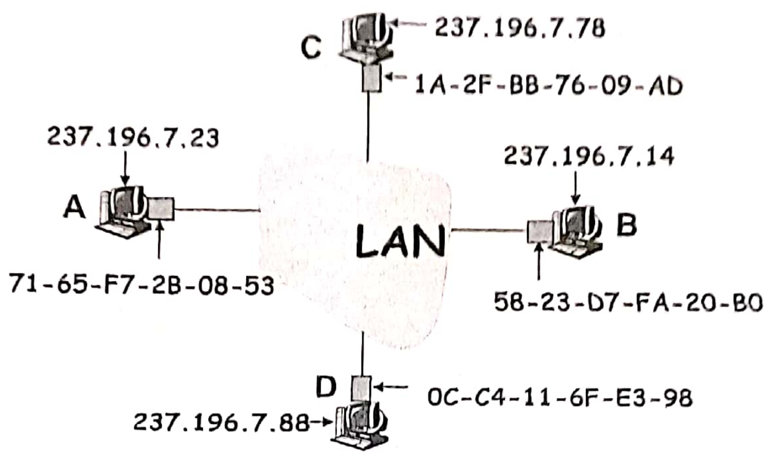
Q 10 __________ is a security measure that involves restricting user access to certain systems, applications, or data based on predefined policies.

Q11 to Q15 Descriptive 3 marks for each question (3x5=15)

Q11 Encrypt the plain text "prejudications" with the key "microprojectors" using Playfair cipher. Also, verify the plain text from the generated cipher text. **03 Marks**

Q 12 Describe the role of the network layer in the OSI model. **03 Marks**

Consider the following Network: **(for Q 13-14)**



Q13 Consider the above network topology, **User A** wants to communicate with **User B.** explain ARP protocol with respect to this scenario. Further consider **User C** as the attacker and explain the ARP spoofing. ***03 Marks***

Q14 With respect to the same network topology, explain TCP *Session Hijacking* and its countermeasures for this network attacks mentioned in question 13. **03 Marks**

Q 15 Explain following examples/terms: (Any two) **03 Marks**

(i) VPN vs VLAN

(ii) Local DNS vs TLD

(iii) IDS vs IPS

National Forensics Sciences University, Goa Campus
Mid- semester Examination

Branch – M.Sc. Cyber Security / M.Tech. AI & DS
Sem – 3
Date - 11-02-2025
Subject Name - Mobile Security / Mobile Security and Forensics
Subject Code - CTMSCS SII P3/ CTMTAIDS SII P2
Duration: 45 Min
Max. Marks- 25
Instructions - **1) Answer all questions. 2) Assume suitable data.**

Q.1	Multiple Choice Questions (1 mark each)	10 marks
	i. Android was developed by the: a. Open Handset Alliance (OHA) b. Close Handset Alliance c. International Android organization d. None of the above	1 mark
	ii. Which of the following is not the part of application framework? a. Content provider b. View System c. Resource Manager d. Surface Manager	1 mark
	iii. What is the main purpose of the Linux Kernel in Android architecture? a. To provide UI component. b. To manage application lifecycle. c. To handle memory management, drivers, and security. d. To provide APIs for app development	1 mark
	iv. Activity Manager lies in which layer of android architecture? a. Kernel b. Libraries c. Application Framework d. Applications	1 mark
	v. Which component of Android architecture provides core system services like telephony and resource management? a. Application Framework. b. Android Runtime. c. Linux Kernel. d. Libraries	1 mark
	vi. Android's native libraries are written in: a. JAVA b. C or C++ c. Python	1 mark

		d. Perl	
	vii.	Which is not an Android Application Components: a. Activity b. Service c. Content d. Assembler	1 mark
	viii.	Which of the following is NOT a part of the Android Application Framework? a. Activity Manager. b. Window Manager. c. SQLite. d. Content Providers	1 mark
	ix.	Which is not a type of intent a. Explicit b. Implicit c. None	1 mark
	x.	What is the role of the Android Runtime (ART) in Android architecture? a. Provides libraries for app development. b. Executes and optimizes Android applications. c. Manages low-level system processes. d. Handles the UI rendering	1 mark
Q.2	Answer any 3 questions (3x5 marks each)		15 Marks
	i.	What is ADB? Explain any five ADB commands.	5 marks
	ii.	What is Android. Explain the architecture of Android with the help of a diagram	5 marks
	iii.	What are different managers in Android Application Framework? Briefly describe any five of them. Not, cp, vs, Act, win resource	5 marks
	iv.	Explain OWASP top 10 vulnerabilities for Mobile.	5 marks

d, r, sh, start, shutdown

National Forensics Sciences University, Goa Campus
Mid-semester Examination

Programme – M.Sc. CS	Sem – II	Date- 17.03.2025
Subject Name: Incident Response & Digital Forensics		Subject Code: CTMSDFJS SI P3
Time- 1.5 Hours		Max.Marks- 50
Instructions - 1) Answer all questions. 2) Assume suitable data.		

Q.1	Solve any four	20 marks
	a. List out the benefits of Incident management.	5 marks
	b. How do you identify the occurrence of an incident?	5 marks
	c. Discuss the different functions/activities of the incident reporting organization such as CERT-IN.	5 marks
	d. How do you survey a digital crime scene?	5 marks
	e. Explain the procedure to handle an incident faced by a manufacturing plant.	5 marks
Q.2	Attempt all	15 marks
	a. List out the different categories of the incident.	5 marks
	b. List out the different tools used in IRM.	5 marks
	c. How do you estimate the cost of an incident?	5 marks
Q. 3	Attempt a and b	15 marks
Q.3 a	Attempt any one	
Q.3 a	I. What is Malware? How do you analyse it using different techniques?	8 marks
	OR	
	II. Explain in detail how a botnet attack occurs with an example.	8 marks
Q.3 b	Attempt any one	7 marks
Q3 b	I. Explain hard drive imaging, how it is different from hard drive cloning.	7 marks
	OR	
	II. Explain the standard operating procedure for cyber forensics.	7 marks

Legal

National Forensics Sciences University, Goa Campus
TA-1 Examination

Program Name – MSC CS/MSC DFIS | Sem – 3 | Date – 11/09/2025

Subject Name - Social Network Analysis | Subject Code- CTMSCS SIII P5/CTMSDFIS SIII P5 PE-2

Time- 45 minutes | Max. Marks- 25

Instructions - 1) Answer all questions.

Q.1 Short Questions (5 mark each) — 10 Marks

A. Discuss about Parties interested in OSINT Information. — 5 marks

OR

A. Explain about Graph Theory. — 5 marks

B. Discuss OSINT Entities and Organisations. — 5 marks

Q.2 Answer any 3 questions — 15 Marks

A. Explain Four categories of open information and intelligence as per NATO literature. — 8 marks

B. What is OSINT, explain in detail. — 7 marks

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National Forensic Sciences University, Goa Campus
Mid-semester Examination

Program – M. Sc. Cyber Security/DFIS	Sem – III	Date- 09.10.25	
Subject Name- Cloud Security & Forensics	Subject Code- CTMSCS S3-P3/CTMSDFIS SIII P4 PE-1		
Time- 1.5 Hours		Max.Marks- 50	
Instructions - 1) Answer all questions. 2) Assume suitable data.			

Q.1	Solve any four	20 marks
	a. Explain block-based storage in cloud computing.	5 marks
	b. With a neat diagram, explain each component of the Xen hypervisor.	5 marks
	c. Explain the concept of virtualization layers.	5 marks
	d. Explain the different components of SOC-2 certification.	5 marks
	e. How does a Kernel-based virtual machine work? Explain.	5 marks
Q.2	Attempt all	15 marks
	a. How object-based storage is different from file-based storage.	5 marks
	b. Write a note on containerization.	5 marks
	c. What is Para virtualization? List out at least three hypervisors based on it.	5 marks
Q. 3	Attempt a and b	15 marks
Q.3 a	Attempt any one	
Q.3 a	I. Differentiate between KVM, VMware ESXi, Hyper-V, and Xen hypervisors.	8 marks
	OR	
	II. Explain PCI passthrough in detail.	8 marks
Q.3 b	Attempt any one	7 marks
Q3 b	I. How do you investigate a case on Amazon Cloud Drive?	7 marks
	OR	
	II. What is a Docker? How containers are different from images; explain with a suitable example.	7 marks

National Forensics Sciences University, Goa Campus
Mid- semester Examination

Programme – M. Sc. Cyber Security   Sem – III   Date – 07-10-2025

Subject Name: Blockchain and Cryptocurrency (CTMSCS SIII P1)

Time- 1.5 Hours   Max. Marks- 50

Instructions - 1) Answer all questions. 2) Assume suitable data.

Q.1	Solve any four	**20 marks**
	a. What are the main objectives of cryptography explain the types.	5 marks
	b. Directed Acyclic Graph (DAG) in Blockchain Technology	5 marks
	c. What are points that need to be consider while designing the Blockchain Network.	5 marks
	d. Explain any two types of Consensus mechanisms in blockchain.	5 marks
	e. How the transaction fees in Bitcoin and Ethereum are calculated.	5 marks
Q.2	Attempt all	**15 marks**
	a. Define a memory-hard algorithm and explain its significance in blockchain and cryptographic systems.	5 marks
	b. Explain the role of Patricia Trees in blockchain technology.	5 marks
	c. Compare the Soft forks and hard forks.	5 marks
Q. 3	a. Draw and explain the methodology of the mini project that you have proposed.	**8 marks**
	OR	
	b. Write the key technical points that you learned in the workshop. Write any one work in detail.	8 marks
Q.4	a. What are the Key Concepts in Mining of Blockchain. Explain.	**7 marks**
	OR	
	b. What is Blockchain governance and Off-chain and On-chain governance. Write a note on Anonymity in blockchain.	(4+3) marks

*** All the best***

National Forensics Sciences University, Goa Campus
Mid- semester Examination

Programme–M.Sc. Cyber Security Date- 08/10/2025 Subject Name: IoT Security and Forensics Time- 1.5 Hours Instructions - 1) Answer all questions. 2) Assume suitable data.		Sem – III Subject Code- CTMSCS SIII P2 Max.Marks- 50
Q.1	**Solve any four**	**20 marks**
	a. What are the Characteristics of IoT	5 marks
	b. Write a short notes on HTTP Basics	5 marks
	c. Explain the COAP	5 marks
	d. Write a short note on Software Defined Networking (SDN).	5 marks
	e. What are the Network Operator Requirements?	5 marks
Q.2	**Attempt all**	**15 marks**
	a. Explain MQTT Architecture Capturing Traffic	5 marks
	b. What are the IoT Applications?	5 marks
	c. Explain XMPP Architecture	5 marks
Q. 3	**Attempt a and b**	**15 marks**
Q.3 a	Attempt any one	
Q.3 a	I. Explain the any two a)UART, b)SPI, c)I2C,d)JTAG	8 marks
	OR	
	II. Explain IoT Levels and Deployment Techniques	8 marks
Q.3 b	Attempt any one	7 marks
Q.3 b	I. Explain the IoT Enabling Technologies.	7 marks
	OR	
	II. Write difference between IoT and M2M	7 marks

*** All the best***

National Forensics Sciences University, Goa Campus
TA-1 Examination

Program Name – M.Sc. Digital Forensics and Information Security	Sem – III
Date- 09/09/2025	
Subject Name- IoT Security and Forensics	Subject Code- CTMSCS SIII P2
Time- 45 minutes	Max. Marks- 25

I	**Short Answer Questions**	**15 marks**
	I. What are the Characteristics of IoT .	2 mark
	II. What are the IoT Applications.	2 mark
	III. What are Challenges in IoT.	2 mark
	IV. Briefly explain the Difference between IoT and M2M.	2 mark
	V. Write a short note on Software Defined Networking (SDN).	2 mark
	VI. Briefly explain the Logical Design of IoT.	2.5 Mark
	VII. What are the Network Operator Requirements?	2.5 Mark
II	Answer any 2 questions (2 x 5 marks each)	**10 Marks**
	i. Explain the IoT Levels and Deployment Techniques.	5 marks
	ii. Explain the IoT Enabling Technologies.	5 marks
	iii. Explain the Simple Network Management Protocol (SNMP).	5 marks

All the best

National Forensics Sciences University, Goa Campus
TA-1 Examination

Program Name – M.Sc DFIS Sem – III Date- 10.09.25
Subject Name- Cloud Security & Forensics Subject Code- CTMSDFIS SIII P4
Time- 45 minutes Max. Marks- 25
Instructions - 1) Answer all questions. 2) Assume suitable data.

1	Among IaaS, PaaS, and SaaS, which is most secure? Justify your answer.	3 marks
2	In the cloud computing environment, which attack can be performed at the application layer and OS?	3 marks
3	What is policy ranking? Explain with a suitable example.	3 marks
4	Discuss the possible vulnerabilities with respect to layers in a cloud environment.	3 marks
5	Discuss and draw the NIST cloud security architecture model.	3 marks
6	Discuss and list out the threats associated only with cloud environment.	5 marks
7	What is multi cloud explain with suitable example and also discuss the security risk associated with it?	5 marks